

Leveraging Agentic AI for Video Pre-Production Planning & Asset Sourcing

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Abstract – Creating videos involves a lot of planning where an user needs to decide the choice of shots, find the right stock clips or music, check whether the licenses are in place and put it all together and integrate it into something usable. Executing those tasks manually is time consuming especially for producing larger contents in the video. This paper describes an agentic AI system that can handle the bulk of this pre-production work automatically by scheduling various tasks within the system through a group of various AI agents. The pipeline works through five core stages that are prompt understanding and refinement, decision reasoning to determine the best sourcing strategy, web scraping for licensed media retrieval, relevance matching against the original request, and asset classification to filter and organize the final output.

Out of the 50 test runs, the system chooses to fetch from the web 42% of cases, 52% with AI generation, and a combined approach for the remaining 6%. Once assets are collected, the system compares the original prompts with the created ones; the average relevance and delivered asset score is 86% and 84% respectively. The health score of the overall system averaged to 62%, with 77% of runs finishing at 60% or higher. The system also includes a chat interface so users can ask for changes, swap out assets, or tweak the output without having to start from scratch.

Keywords: *agentic AI, multi-agent systems, video pre-production, asset sourcing, Large Language Models (LLMs), automated planning, content creation, licensing compliance, generative AI, media fetching algorithm.*

1. Introduction

Recent advances in artificial intelligence have undeniably reshaped major industries like education, healthcare, and entertainment. However, building and deploying these systems is still far from easy. For many creators, the true potential of AI is often hidden behind steep technical challenges. It usually requires specialized knowledge in areas like model integration, complex data handling, and multimodal processing. These demands create significant entry barriers, especially for independent creators and smaller organizations that may not have access to large-scale infrastructure or deep technical expertise. As a result, even though generative AI capabilities are rapidly growing, they often remain disconnected and difficult to combine into a single, practical solution. This creates a clear gap between what the technology can do and how easily people can actually use it.

When we look specifically at digital content creation, the pre-production stage stands out as one of the most challenging and time-consuming parts of the process. Despite all the technological progress, this phase still relies heavily on manual effort. Creators often have to plan everything from scratch, gather assets from different sources, and coordinate with multiple stakeholders. This fragmented workflow not only increases production costs but also leads to delays in timelines. On top of that, there are serious legal risks involved. Licensing verification is often done manually, which makes it prone to errors, something that can result in major compliance issues if even one asset, like a stock video or audio clip, is used incorrectly. Most existing tools don't solve this broader problem, instead, they focus on generating individual elements like a single image or a

short video. They lack the ability to handle the bigger picture which are understanding creative intent, organizing assets, and ensuring proper usage rights. Because of this, creators are forced to spend more time managing technical and administrative challenges rather than focusing on what truly matters: telling compelling stories.

2. Literature Review

Recent advances in generative AI have widely transformed content creation, particularly in video and film production, enabling automated synthesis of visual media from text prompt [1-3]. Surveys highlight the rapid progression in generative models for film creation and cinematography [3]. Systems like FilMaster bridge professional filmmaking principles such as camera language and narrative rhythm with generative AI to produce automated, editable films [1]. Similarly, collaborative multi-agent frameworks have emerged to copy human creative processes in content generation. For instance, CREA uses specialized agents to iteratively conceptualize, generate, study and refine outputs using diffusion models, achieving higher novelty and coherence in images and videos [2]. These approaches demonstrate the potential of agentic collaboration for creative tasks, drawing on LLM orchestrated multiagent system [19].

Earlier works laid the foundation for multimodal capabilities, which were essential for video understanding and generation. Models like Flamingo [11] and Kosmos-2 [13] advanced visual language alignment techniques [12] enabled audio visual instruction tuning crossmodel reasoning critical for asset evaluation [10,12]. Unified video language pretraining and binding mechanisms in shared embedding spaces, such as ImageBind [15] and Unibind [16], support consistent cross model representations for tasks like relevance matching [15,16,18]. This foundation helps in decision making strategies as well as semantic filtering in modern pipelines.

Despite these things in media synthesis, Pre-production is not explored that well. Professional video workflow demands structured planning, including intent extraction, shot sequencing, asset governance and licensing compliance, which current generative tools are not taking into account [4-6]. Systematic reviews and impact analyses reveal that AI in digital video production and filmmaking is often focusing on isolated synthesis rather than an end to end orchestrated system, leaving manual coordination for asset collection, verification and integration [4-6]. Tools emphasizing collaborative planning such as VideoDirectorGPT, achieve consistent multi scene video using LLM guided scripts but lack integrated asset sourcing and also branching between licensed retrieval and generation [8]. Long form video agents like VedioAgent are excellent in understanding [9], yet few address proactive pre-production logistics, such as cost latency aware routing or license-safe web scraping from providers versus on-demand generation.

Existing platforms for quick clip generation operate in silos, they are also lacking higher-level features like automated decision reasoning, semantic JSON artifacts for workflow continuity, relevance boosting with metadata, or repetitive feedback loops [7]. Human and AI codesign framework highlights nonlinear collaboration benefits in creative domains [7], but applications to video pre-production balance realism scoring. Feasibility judgement, and conversational refinement remain fragmented. Industry disruption from AI in filmmaking underscores the need for a unified, compliant system that democratizes access while reducing manual work [5,6].

Frameworks like Jockey use planner to worker to reflector architectures to orchestrate specialized video native models for semantic search and understanding, distributed tasks across agents for complex querying and reasoning. In film specific contexts, FilmAgent simulates a virtual production crew which includes director, screenwriter, actors and cinematographers collaborating iteratively in 3D environments to automate script refinement, shot planning and feedback loops. They also help in reducing the hallucinations through structured critique. Such approaches extend human-like workflows into virtual sandboxes, this enables end to end automation that mirrors professional sets. Similarly, extensions of collaborative frameworks like CREA show us a demo of multi agent iteration for creative video generation, where agents conceptualize, critique, and refine outputs to achieve higher artistic novelty [2]. These advancements highlight the value of specialized roles and shared artifacts in maintaining narrative consistency elements that our system incorporates.

Despite these promising multi agent evolutions and advancements in synthesis and virtual production. Practical pre-production challenges such as intelligent routing between licensed stock retrieval and generative synthesis, metadata driven relevance boosting and built in compliance for asset licensing remain largely unaddressed in integrated systems. Industry tools and prototypes often prioritize post synthesis editing or isolated generation. Compliance mechanisms in broader content agents exist in fragmented forms but are rarely embedded in video specific pipelines with decision aware branching.

In summary, while generative AI has advanced in creative synthesis [1-3,8,19,20] , a clear gap persists in practical , production oriented pre-production. Integrating prompt refinement , intelligent sourcing , relevance classification , and user driven iteration into a single agentic pipeline.

3. Methodology

The proposed system is a multi-agent pipeline in which each agent handles different stages in the asset sourcing workflow. A user's creative prompt enters the system and gets converted into a rich and a structured semantic representation. Later a routing decision is made and finally a curated set of classified assets are returned to the user. The system is built in such a way that each agent produces a well structured output which is given as input to the other agents. This reduces ambiguity and hallucination by the agent allowing it to give consistent responses across varied prompts.

3.1 System Architecture

The overall architecture of the pipeline as illustrated in Fig. 1, consists of five sequential stages: Prompt Understanding, Decision Reasoning, Asset Acquisition i.e web scraping or invoking generative models, Relevance Matching and Asset Classification. Rather than treating these modules as independent, there is a shared JSON artifact that is passed across stages which ensures that the context from the prompt analysis is accessible throughout the entire workflow.

The key stage in the pipeline is the branching that occurs after Decision Reasoning. Depending on the nature of the prompt, the pipeline may be routed to either scraping of images or videos from media providers or generation of the assets from scratch using generative models. Both of these stages converge into relevance matching stages which checks for the relevance of the retrieved asset with the user's requirement. The assets are classified in detail and delivered to the user in the end.

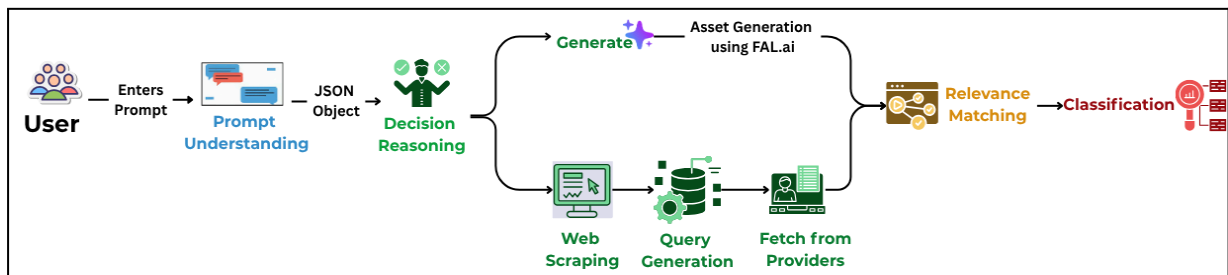


Fig. 1 AutoGenie Agentic AI Pipeline

3.2 Prompt Understanding Agent

The pipeline begins when a user submits a prompt and the prompt understanding agent leverages LLM to understand the prompt in detail. This module evaluates three dimensions for each prompt which includes intent extraction, realism evaluation and feasibility judgment. In intent extraction the LLM extracts several key fields from the raw user prompt such as modality, the primary subject, context scene and style adjectives.

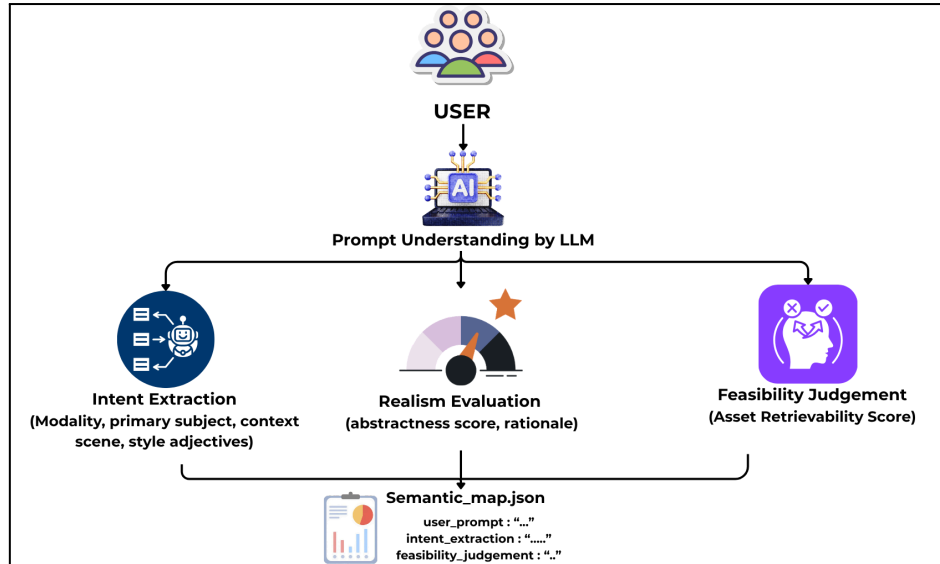


Fig. 2a User Prompt Understanding Agent

In Realism evaluation, an abstractness scoring is done based on the context and the style adjectives to determine if the user's requirement is realistic or imaginary. This scoring is supported by a rationale which states the LLM's reasoning for the scoring it did on the prompt. Finally feasibility judgement is done where asset retrievability scoring is done which determines whether we can find the asset from the media providers. All three branches converge to build a semantic JSON object which is later accessed by the Decision Reasoning agent. Fig. 3 shows the output generated by the Prompt Understanding Agent for the sample prompt: '3 videos of a whimsical boat gently floating among fluffy clouds in a bright blue sky'.

```

{
  "intent_extraction": {
    "modality": "video",
    "domain": "creative arts",
    "primary_subject": "whimsical boat",
    "context_scene": "floating among fluffy clouds in a bright blue sky",
    "style_adjectives": ["whimsical", "imaginative", "playful", "dreamy", "surreal"]
  },
  "realism_scoring": {
    "realism_score": 0.2,
    "abstractness_score": 0.9,
    "rationale": "The scene described is highly imaginative and surreal, depicting a boat floating in the sky among clouds, which is not realistic but rather abstract and whimsical."
  },
  "market_availability_estimate": 0.3
}
  
```

Fig. 2b Output of Prompt Understanding Agent for a whimsical prompt

3.3 Decision Reasoning Agent

The pivotal decision and routing stage is illustrated in Fig.4a. The JSON artifact which includes the intent, the realism and abstraction scoring and the asset retrievability scoring is used by the Decision Reasoning agent. Here LLM is leveraged to estimate the cost and the latency for web scraping or generation using models in accordance to the user's requirements. If the cost and the latency through web scraping is high as compared to generation then the orchestrator calls the Generation agent instead of Web scraping agent and vice versa.

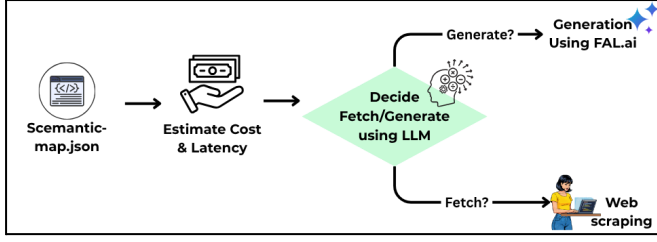


Fig. 3a Decision Reasoning Agent

```

--- DECISION REASONING MODULE ---
Low realism (fantasy/abstract) -> generate.
Decision: generate_with_model
  
```

Fig. 3b Decision Reasoning Agent routing to Generation Agent

This cost-aware mechanism ensures that the system selects most appropriate, economical, and high-quality path balancing performance, speed, and output fidelity before advancing to asset production. Fig. 4b shows the output generated by the Decision Reasoning Agent for the same prompt. Since the prompt mentions a whimsical boat floating through fluffy clouds which is not realistically possible, the agent decides to generate this asset.

3.4 Web Scraping Agent

The detailed workflow of the Web Scraping Agent is shown in Fig. 4a, responsible for efficiently retrieving existing media assets when the decision engine routes a high-realism prompt to fetching rather than generation. It begins with Query Expansion using LLM to transform the original user prompt into multiple semantically rich search queries. These queries improve recall and relevance across diverse stock providers. The queries are then fed into a Provider Routing module that intelligently selects the most appropriate sources to fetch the asset based on quality requirements and content type. This is followed by media fetching which fetches the candidate assets from unsplash, pexels, pixabay etc. After this the media is extracted along with their metadata and other technical aspects such as size, ratio, file format alongside source level descriptors. Before storage, the candidate assets undergo deduplication where a perpetual hashing is used to find near identical assets and they are removed from the pool.

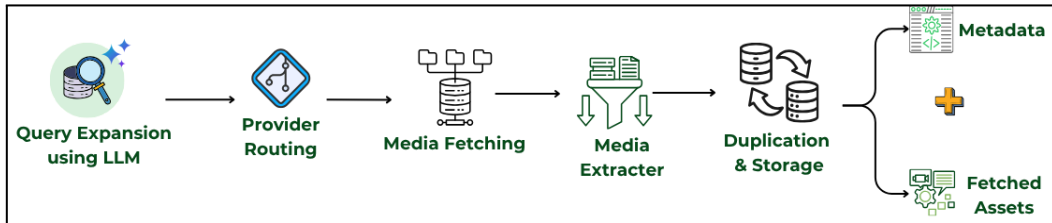


Fig. 4a Web Scraping Agent

```

[route] modality detected: image
[llm] queries: [ 'barista pouring latte art', 'coffee shop barista close-up' ]
[scrape] unsplash -> 'barista pouring latte art'
[scrape] pexels -> 'barista pouring latte art'
[scrape] pixabay -> 'barista pouring latte art'
[scrape] unsplash -> 'coffee shop barista close-up'
[scrape] pexels -> 'coffee shop barista close-up'
[scrape] pixabay -> 'coffee shop barista close-up'
[scrape] total collected items: 15
  
```

Fig. 4b Media results collected by Web Scraping Agent

The final output, as shown in Fig. 4b for the prompt “A highly detailed cinematic video of a man confidently delivering a speech on a well-lit stage” shows how each media provider is scraped using the generated LLM queries and thus 15 of them are scraped as candidates and sent to the Relevance Matching Agent.

3.5 Generation Agent

When the Decision Reasoning Agent routes a prompt to the generation path, the Generation Agent is responsible for synthesizing new media from scratch. It takes the JSON artifact as input and begins with creating an enriched query that combines the primary scene, scene description, domain etc to create a highly detailed prompt string. This constructed prompt is more descriptive than the user's original prompt which helps to generate assets that closely match the user's requirements. Once the query is generated a model selection step identifies the appropriate model from [FAL.ai](https://fal.ai) based on the requested modality. The asset is generated by the model using the enriched prompt and stored locally along with the associated metadata, media URL and the query string used.

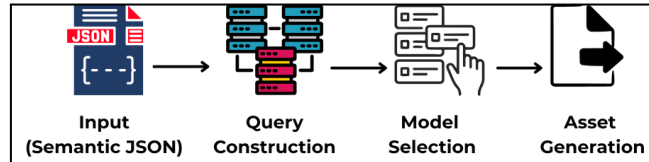


Fig. 5a Generation Agent

```

[generation] modality detected: video
[generation] base prompt for Fal:
Create 3 videos of a whimsical boat gently floating among fluffy clouds in a bright blue sky. The boat should have an imaginative and playful design, such as a sailboat with colorful sails or a steampunk airship. The clouds should appear soft and cotton-like, and the sky should be a vibrant blue, creating a dreamy and surreal atmosphere. | intent: creative arts, whimsical boat, floating among fluffy clouds in a bright blue sky, whimsical, imaginative, playful, dreamy, surreal, cinematic, high quality video
[generation] triggering 3 parallel video generations...
[generation] Fal returned 3 video URLs
[generation] downloading video: https://v3b.fal.media/files/b/0a935a1b/eGX5T9_w566GFQpA63IKP_0yBAIrFW.mp4
[generation] downloading video: https://v3b.fal.media/files/b/0a935a1b/z1LtcNkmO3viCLUfnYRjr_MP2w0GXs.mp4
[generation] downloading video: https://v3b.fal.media/files/b/0a935a1b/qCpnvkOJMbAKx3pXRIM-c_SQL1ousa.mp4
[generation] stored 3 generated video assets. See scrape_assets\metadata.json
  
```

Fig. 5b Results of the Generation Agent

Fig.5b shows the refined prompt before triggering of video generation. The approach of separating query construction from model invocation proved useful during testing, as it allowed us to adjust the enrichment logic independently without modifying the generation call itself.

3.6 Relevance Matching Agent

Regardless of whether the assets are fetched or generated, all candidates pass through a Relevance Matching agent before classification. This agent computes a similarity score between the user's original prompt and each candidate asset using CLIP embeddings. CLIP encodes both the asset and the prompt in a shared representation space and thus a cosine similarity between the prompt embedding and each asset's embedding produces the initial relevance score.

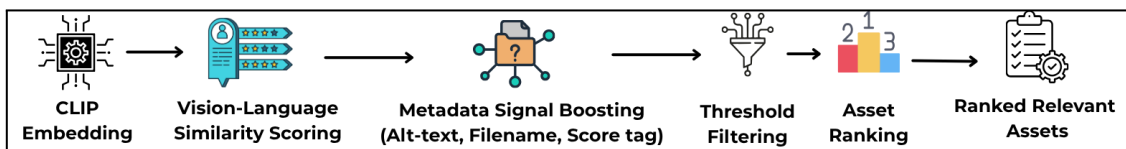


Fig. 6a Relevance Matching Agent

This score is boosted using metadata signals such as alternative text corresponding to the asset from the media provider, filename relevance to produce a composite relevance measure. A configurable threshold filter then removes the low scoring candidates and ranks the selected assets on the basis of the final composite score. Fig. 5b shows the final composite scores of the three generated videos as 95%, 96% and 97%. All of these three videos are ranked and retained for classification.

```

=====
INTENT-AWARE RELEVANCE MATCHING
=====
[relevance] fetched_assets count: 3
Intent Extracted: {
  primary_subject: 'whimsical boat',
  primary_action: 'create',
  supporting_objects: [
    '3 videos',
    'fluffy clouds',
    'bright blue sky',
    'imaginative and playful design',
    'colorful sails',
    'steampunk airship',
    'soft cotton-like clouds',
    'vibrant blue sky'
  ],
  relationships: [ 'among fluffy clouds' ],
  conflicting_actions_to_reject: []
}
Evaluating 9 images
✓ Selected (0.95): fal-ai-wan-v2.2-a14b-text-to-video_48f212c5868b.mp4
✓ Selected (0.97): fal-ai-wan-v2.2-a14b-text-to-video_bfd7c200474f.mp4
✓ Selected (0.96): fal-ai-wan-v2.2-a14b-text-to-video_6044d41603f5.mp4
Finished. Selected: 3

```

Fig. 6b Relevance scoring done for each candidate asset

3.7 Asset Classification Agent

The final stage of the pipeline annotated each retained asset with a structured profile. Video assets undergo frame extraction for detailed classification while images are processed directly. Both types are then analyzed by Open AI's gpt-4o model that identifies scene type, environment, time of day, lighting conditions, mood, shot type etc. This ensures that the user is well presented both the assets and well categorized content that aligns closely with the original user intent across diverse prompt types.

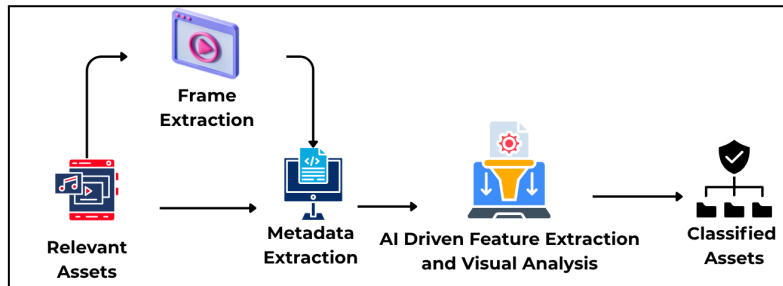


Fig. 7a Asset Classification Agent

Fig. 6b shows the result of the asset classification agent for the videos of boats floating in clouds which will help the user to effectively use the assets for their preferred use case.

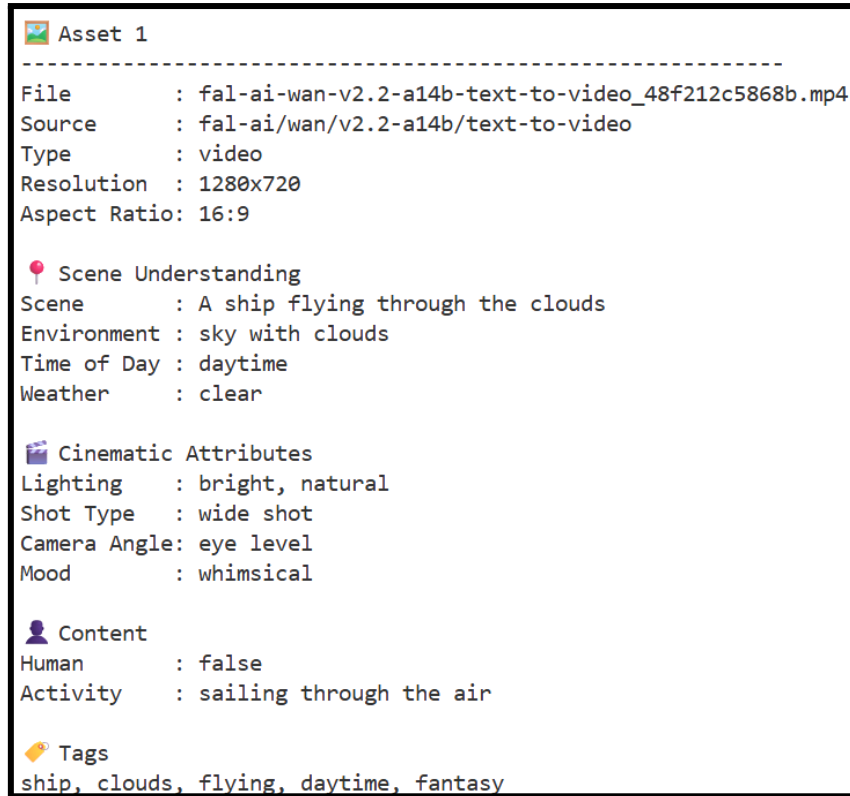


Fig. 7b Final classification of the chosen asset

If the assets returned by either the Web Scraping Agent or the Generation Agent fail to meet the relevance threshold, the system initiates a feedback loop. In this loop, the Decision Reasoning Agent re-examines the cause of the poor match and adjusts its routing decision by either switching strategies or refining the query before attempting a second retrieval or generation pass.

4. Results and Discussions

Test Case	Prompt 1: “3 videos of a whimsical boat gently floating among fluffy clouds in a bright blue sky”	Prompt 2: “An image of an anthropomorphic cat dressed in full medieval knight armor, standing valiantly on a battlefield with a dramatic sky and scattered debris around.”	Prompt 3: “A skilled barista pouring intricate latte art into a white ceramic cup, with a focus on the swirling patterns of the milk and coffee.”
Result Images	Image captured out of video of boats flying on the clouds	Image of a cat getting ready for war against its enemies	A Barista showing his art work to the people

			
Prompt Understanding Summary	• Modality: video • Requested count: 3 • Primary subject: whimsical cartoon-style boat • Scene: gently floating among fluffy white clouds in bright blue sky • Style: whimsical, cartoon style	• Modality: image • Requested count: 1 • Primary subject: anthropomorphic cat • Context scene: battlefield • Style adjectives: dramatic, valiant, medieval • Domain: fantasy • Full intent: fantasy scene of valiant armored cat warrior on battlefield with dramatic sky & debris	• Modality: image • Count: 1 • Subject: barista pouring latte art • Scene: coffee prep • Style: intricate, swirling • Domain: food & beverage • Intent: realistic close-up of latte art in white cup
Decision Reasoning & Scoring	• High cartoon / abstraction: direct animation/video gen • Low realism need • Agentic confidence: 0.9 • Rationale: fantastical scene suits generative diffusion models • Total Latency: ~29s	• Reasoning trace: "Low realism indicates fantasy or abstract content. Generating." • Final decision: generate_with_model • Confidence: 0.85 • Stage metrics: completeness_score=1 (stage1), best_match_score=0.97 (stage4), modality_confidence=0.9 • Total latency: ~10s	• Decision: fetch_from_web • Confidence: 0.85 • Realism high → web fetch chosen • Metrics: completeness=1, best_match=0.98, filtering≈0.56 • Latency: ~75s (web-heavy)
Web Scraping / Search Performed?	Minimal / none (semantic/internal search sufficient)	Yes – Generated via fal-ai/flux/dev model (3 images produced with near-identical query) Query used: "An image of an anthropomorphic cat dressed in full medieval knight armor, standing valiantly on a battlefield with a dramatic sky and scattered debris around. intent: fantasy, anthropomorphic cat, battlefield, dramatic, valiant, medieval" No external web/stock scraping; direct model generation	Yes – Unsplash, Pexels, Pixabay • Queries: "barista latte art coffee shop" + "coffee preparation skilled barista" • Many candidates fetched, 3 providers

Relevance Matching Result	• High relevance to cartoon sky + boat assets • Strong match for dreamy/surreal floating motifs • Perspectives via multi-angle prompts	• Very high relevance (precision_at_k=1, mrr=1) • All fetched/generated assets match prompt closely • Best match score: 0.97 • Visual diversity score: 0 (variations are subtle: sunset vs day, cloudy vs clear, slight palette/pose differences)	• Very high: precision=1, mrr=1, best_match=0.98 • Strong matches: milk pouring, swirling patterns, white cups • Diversity: low (similar close-ups)
Final Asset Classification & Outcome	• Asset type: video (cartoon/animation) • Count: 3 • Style: whimsical cartoon • isComplete: true • Output: 3 short looping clips	• Asset type: image (generated) • Count: 3 generated (even though requested 1; pipeline over-generated for selection) • Source: fal-ai/flux/dev • isComplete: true (high confidence generation) • Key classifications (common across assets): - Primary scene: Anthropomorphic cat in armor / warrior - Environment: Medieval fantasy battlefield / open rocky landscape - Mood/atmosphere: Heroic, epic - Shot type: Full-body / full shot - Lighting: Dramatic / soft natural - Dominant objects: cat, armor, sword, cape - Tags: fantasy, cat, knight, armor, medieval, epic, battlefield, sunset/cloudy • All assets: 1024×768, 4:3 aspect, confidence ~0.95	• Type: scraped/stock image • Sources: mainly Unsplash & Pexels • isComplete: true • Common traits: - Scene: pouring latte art - Shot: close-up / top-down - Mood: calm, cozy - Objects: white cup, milk pitcher, hands - Tags: latte art, barista, café • Confidence: 0.9–0.95
Realism / Abstraction Score	• Realism: 0.1 • Abstraction: 0.9	• Realism: 0.2 • Abstraction: 0.8 • Rationale: "The concept of an anthropomorphic cat in knight armor is highly fantastical, making it more abstract than realistic."	• Realism: 0.9 • Abstraction: 0.1 • Rationale: Common real-world coffee activity
Discussion	The pipeline detected very low realism (0.1) and high abstraction (0.9) in the whimsical boat prompt, correctly routing it straight to video generation instead of web search, as no real stock footage exists for a cartoon boat floating in clouds.	The pipeline detected low realism (0.2) and high abstraction (0.8) in the anthropomorphic cat knight prompt, correctly routing it to direct AI generation with fal-ai/flux/dev instead of web search, as market availability was only 0.3 for such a specific fantasy concept.	The pipeline identified high realism (0.9) and low abstraction (0.1) in the barista latte art prompt, correctly choosing web fetch over generation since market availability was solid at 0.7 for this popular coffee culture topic with abundant high-quality stock photos from Unsplash, Pexels, and Pixabay.

Table no. 1 Results & Discussions

5. Conclusion

This paper presented a new system for automating the video pre-production process using a multi-agent architecture coordinated by large language models. The system takes a user's text prompt and works through it end to end breaking it down into structured requirements, deciding whether to fetch existing licensed media or generate new assets, retrieving and scoring candidates for relevance, and presenting the final results through a conversational interface. Each of these stages is handled by a dedicated agent, which keeps the pipeline modular and makes individual components easier to test and improve independently.

A key strength of this approach is in bringing together tasks that are normally scattered across different tools and manual steps like prompt interpretation, asset sourcing, license-safe retrieval, semantic filtering into one unified pipeline. The conversational interface further allows users to iterate on results without restarting, which saves time during revisions. Together, these features make professional-level pre-production planning accessible to a much wider range of users, from independent creators to production teams.

As future work, there is room to improve how the system handles artistic intent and tonal context, since text-visual matching through CLIP embeddings does not always capture the mood a user has in mind. Reducing dependency on external API response times and exploring ways to better handle edge cases in LLM-based prompt analysis are also areas worth investigating. These refinements would further strengthen the system's reliability and creative output quality.

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Table 1. Architectural Diagrams

Sr. No.	Architectural Diagrams	
	Heading	Page No.
1.	AutoGenie Agentic AI Pipeline	3
2.	a. User Prompt Understanding Agent	4
	b. Output of Prompt Understanding Agent for a whimsical prompt	4
3.	a. Decision Reasoning Agent	5
	b. Decision Reasoning Agent routing to Generation Agent	5
4.	a. Web Scraping Agent	5
	b. Media results collected by Web Scraping Agent	5
5.	a. Generation Agent	6
	b. Results of Generation Agent	6
6.	a. Relevance Matching Agent	6
	b. Relevance scoring done for each candidate asset	7
7.	a. Asset Classification Agent	7
	b. Final classification of the chosen asset	8

Table 2. List of Tables

Sr. No.	List of Tables	
	Heading	Page No.
1.	Results & Discussions	8